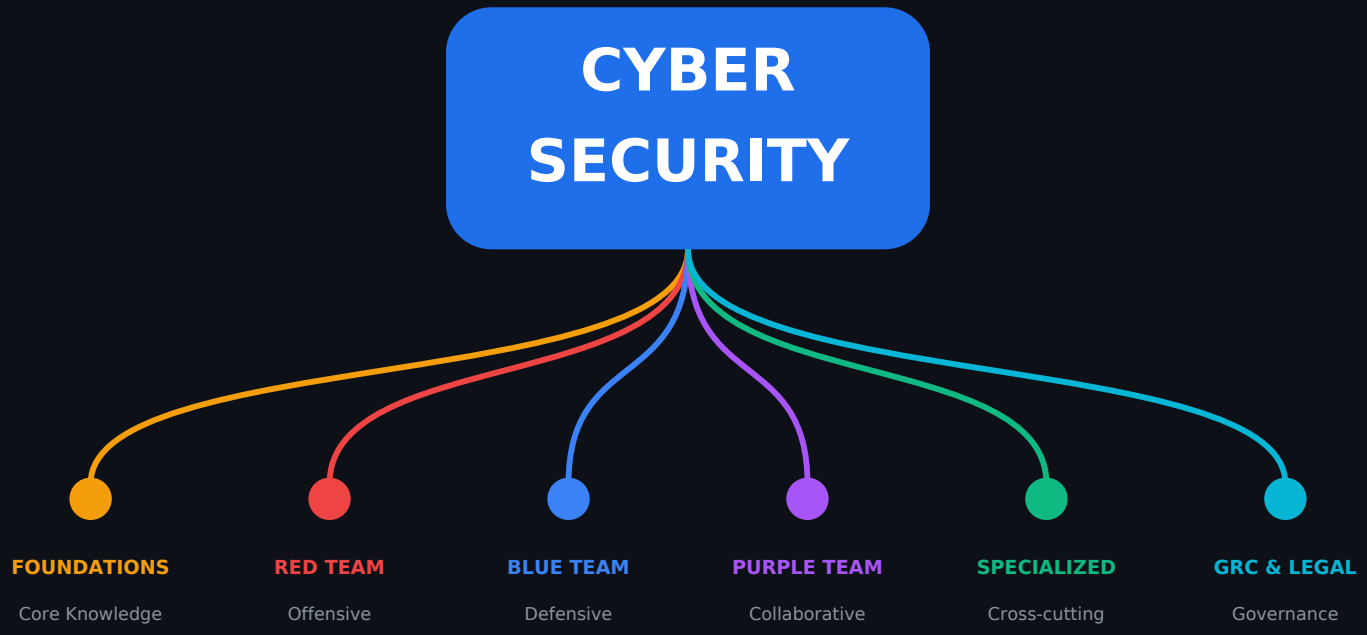
The Cyber Security Tree

Red Team | Blue Team | Purple Team | Every field, tool & link



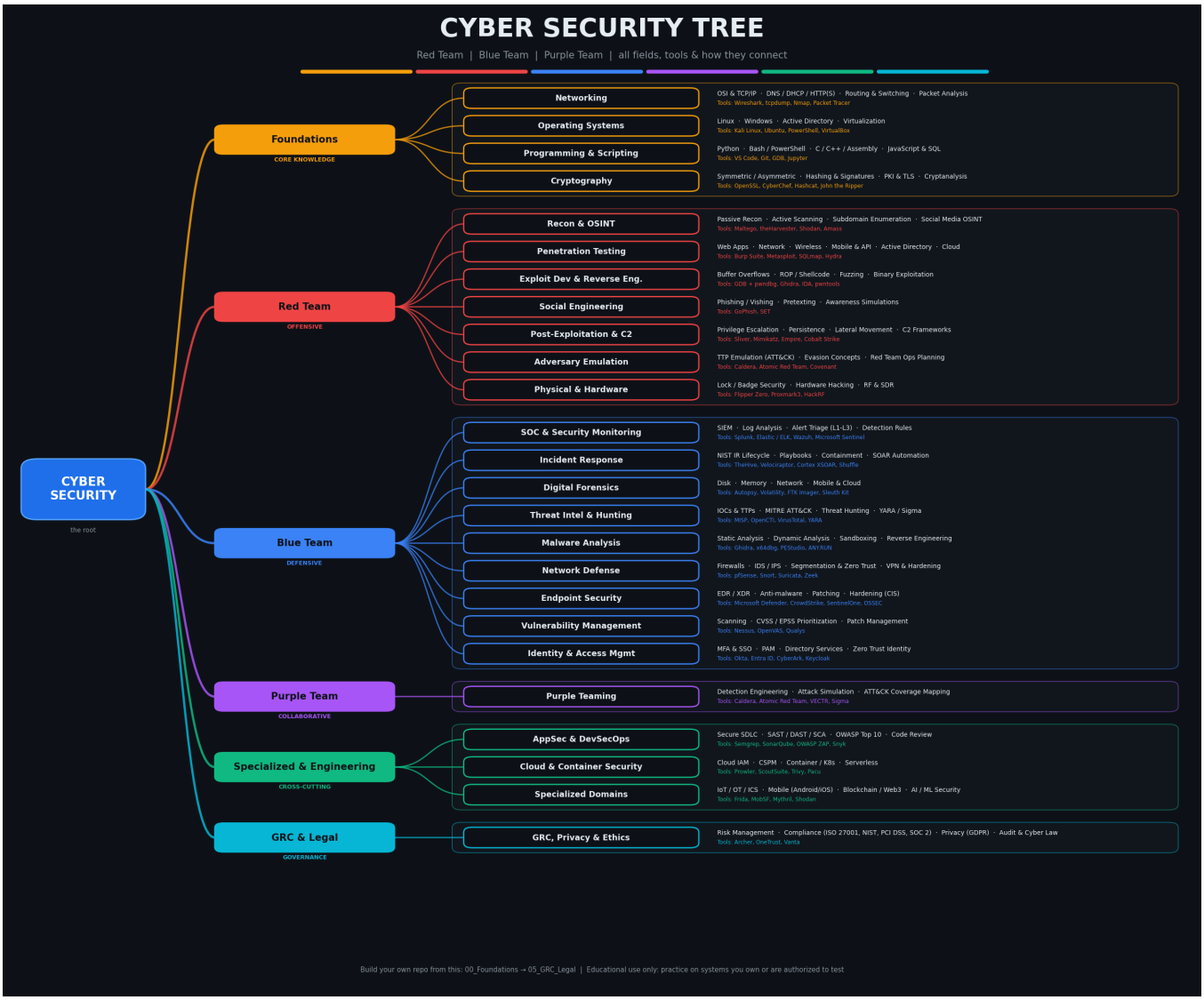
6	25	101	36
branches	fields	tools	links

Educational reference. Only test systems you own or have written permission to test.

github.com/Syed-Muhammad-Tayyab

1. The Full Tree

Root, 6 branches and 25 fields with their sub-fields and tools.



Branch	Type	What it does	Fields
[ ] **Foundations**	Core Knowledge	The base every security role is built on	4
[ ] **Red Team**	Offensive	Think like the attacker: break in, prove impact	7
[ ] **Blue Team**	Defensive	Detect, respond, defend and recover	9
[ ] **Purple Team**	Collaborative	Red and Blue working together to improve detection	1
[ ] **Specialized & Engineering**	Cross-cutting	Secure-by-design fields that serve both sides	3
[ ] **GRC & Legal**	Governance	Policies, risk, compliance, audit and ethics	1

2. Foundations

Core Knowledge | The base every security role is built on

Networking `00_Foundations/Networking/`

How data moves: the language every attacker and defender speaks.

Sub-fields	OSI & TCP/IP | DNS / DHCP / HTTP(S) | Routing & Switching | Packet Analysis
Tools	Wireshark, tcpdump, Nmap, Packet Tracer
Certs	CCNA, Network+

Operating Systems `00_Foundations/Operating_Systems/`

Linux, Windows and Active Directory internals.

Sub-fields	Linux | Windows | Active Directory | Virtualization
Tools	Kali Linux, Ubuntu, PowerShell, VirtualBox
Certs	Linux+, LPIC-1

Programming & Scripting `00_Foundations/Programming_Scripting/`

Automate, analyze and build security tooling.

Sub-fields	Python | Bash / PowerShell | C / C++ / Assembly | JavaScript & SQL
Tools	VS Code, Git, GDB, Jupyter
Certs	PCAP

Cryptography `00_Foundations/Cryptography/`

Protect confidentiality, integrity and authenticity.

Sub-fields	Symmetric / Asymmetric | Hashing & Signatures | PKI & TLS | Cryptanalysis
Tools	OpenSSL, CyberChef, Hashcat, John the Ripper
Certs	Crypto I (Coursera)

3. Red Team

Offensive | Think like the attacker: break in, prove impact

Recon & OSINT `01_Red_Team/Recon_OSINT/`

Map the target's attack surface before touching it.

Sub-fields	Passive Recon | Active Scanning | Subdomain Enumeration | Social Media OSINT
Tools	Maltego, theHarvester, Shodan, Amass, Recon-ng
Certs	TCM OSINT

Penetration Testing `01_Red_Team/Penetration_Testing/`

Authorized exploitation of web, network and more.

Sub-fields	Web Apps | Network | Wireless | Mobile & API | Active Directory | Cloud
Tools	Burp Suite, Metasploit, SQLmap, Hydra, BloodHound
Certs	eJPT, PNPT, OSCP

Exploit Dev & Reverse Eng. `01_Red_Team/Exploit_Development/`

Find memory bugs and turn them into working exploits.

Sub-fields	Buffer Overflows | ROP / Shellcode | Fuzzing | Binary Exploitation
Tools	GDB + pwndbg, Ghidra, IDA, pwntools, AFL++
Certs	OSED, OSEE

Social Engineering `01_Red_Team/Social_Engineering/`

Exploit people, the most common entry point.

Sub-fields	Phishing / Vishing | Pretexting | Awareness Simulations
Tools	GoPhish, SET
Certs	SEPP

Post-Exploitation & C2 `01_Red_Team/Post_Exploitation_C2/`

Escalate, persist, move laterally and control access.

Sub-fields	Privilege Escalation | Persistence | Lateral Movement | C2 Frameworks
Tools	Sliver, Mimikatz, Empire, Cobalt Strike
Certs	CRTO, CRTP

Adversary Emulation `01_Red_Team/Adversary_Emulation/`

Full red-team operations that copy real threat actors.

Sub-fields	TTP Emulation (ATT&CK) | Evasion Concepts | Red Team Ops Planning
Tools	Caldera, Atomic Red Team, Covenant
Certs	CRTO, CRTL

Physical & Hardware `01_Red_Team/Physical_Hardware/`

Doors, badges, devices and embedded systems.

Sub-fields	Lock / Badge Security | Hardware Hacking | RF & SDR
Tools	Flipper Zero, Proxmark3, HackRF
Certs	TCM Physical

4. Blue Team

Defensive | Detect, respond, defend and recover

SOC & Security Monitoring `02_Blue_Team/SOC_Monitoring/`

24/7 eyes on logs and alerts.

Sub-fields	SIEM | Log Analysis | Alert Triage (L1-L3) | Detection Rules
Tools	Splunk, Elastic / ELK, Wazuh, Microsoft Sentinel, QRadar
Certs	Security+, BTL1, SC-200

Incident Response `02_Blue_Team/Incident_Response/`

Contain, eradicate and recover from attacks.

Sub-fields	NIST IR Lifecycle | Playbooks | Containment | SOAR Automation
Tools	TheHive, Velociraptor, Cortex XSOAR, Shuffle
Certs	GCIH, ECIH

Digital Forensics `02_Blue_Team/Digital_Forensics/`

Preserve and analyze evidence.

Sub-fields	Disk | Memory | Network | Mobile & Cloud
Tools	Autopsy, Volatility, FTK Imager, Sleuth Kit
Certs	GCFE, GCFA, CHFI

Threat Intel & Hunting `02_Blue_Team/Threat_Intel_Hunting/`

Know the adversary, hunt before alerts fire.

Sub-fields	IOCs & TTPs | MITRE ATT&CK | Threat Hunting | YARA / Sigma
Tools	MISP, OpenCTI, VirusTotal, YARA, Sigma
Certs	GCTI, CTIA

Malware Analysis `02_Blue_Team/Malware_Analysis/`

Dissect malicious code to understand and block it.

Sub-fields	Static Analysis | Dynamic Analysis | Sandboxing | Reverse Engineering
Tools	Ghidra, x64dbg, PEStudio, ANY.RUN, Cuckoo
Certs	GREM, eCMAP

Network Defense `02_Blue_Team/Network_Defense/`

Build networks that resist and reveal attacks.

Sub-fields	Firewalls | IDS / IPS | Segmentation & Zero Trust | VPN & Hardening
Tools	pfSense, Snort, Suricata, Zeek, Fortinet
Certs	CCNA Security, NSE 4

Endpoint Security `02_Blue_Team/Endpoint_Security/`

Protect laptops, servers and phones.

Sub-fields	EDR / XDR | Anti-malware | Patching | Hardening (CIS)
Tools	Microsoft Defender, CrowdStrike, SentinelOne, OSSEC
Certs	CySA+

Vulnerability Management `02_Blue_Team/Vulnerability_Management/`

Find, rank and fix weaknesses continuously.

Sub-fields	Scanning | CVSS / EPSS Prioritization | Patch Management
Tools	Nessus, OpenVAS, Qualys
Certs	CySA+

Identity & Access Mgmt `02_Blue_Team/Identity_Access/`

Right person, right access, right time.

Sub-fields	MFA & SSO | PAM | Directory Services | Zero Trust Identity
Tools	Okta, Entra ID, CyberArk, Keycloak
Certs	SC-300, CISSP (IAM)

5. Purple Team

Collaborative | Red and Blue working together to improve detection

Purple Teaming `03_Purple_Team/Purple_Teaming/`

Attack, detect, tune, repeat.

Sub-fields	Detection Engineering | Attack Simulation | ATT&CK Coverage Mapping
Tools	Caldera, Atomic Red Team, VECTR, Sigma
Certs	SANS SEC599

6. Specialized & Engineering

Cross-cutting | Secure-by-design fields that serve both sides

AppSec & DevSecOps `04_Specialized_Engineering/AppSec_DevSecOps/`

Build security into the software lifecycle.

Sub-fields	Secure SDLC | SAST / DAST / SCA | OWASP Top 10 | Code Review
Tools	Semgrep, SonarQube, OWASP ZAP, Snyk
Certs	CSSLP, OSWE

Cloud & Container Security `04_Specialized_Engineering/Cloud_Security/`

AWS, Azure, GCP, Docker and Kubernetes.

Sub-fields	Cloud IAM | CSPM | Container / K8s | Serverless
Tools	Prowler, ScoutSuite, Trivy, Pacu
Certs	AZ-500, CCSP, AWS Security

Specialized Domains `04_Specialized_Engineering/Specialized_Domains/`

IoT, OT/ICS, mobile, blockchain and AI security.

Sub-fields	IoT / OT / ICS | Mobile (Android/iOS) | Blockchain / Web3 | AI / ML Security
Tools	Frida, MobSF, Mythril, Shodan
Certs	GICSP, OSMR

7. GRC & Legal

Governance | Policies, risk, compliance, audit and ethics

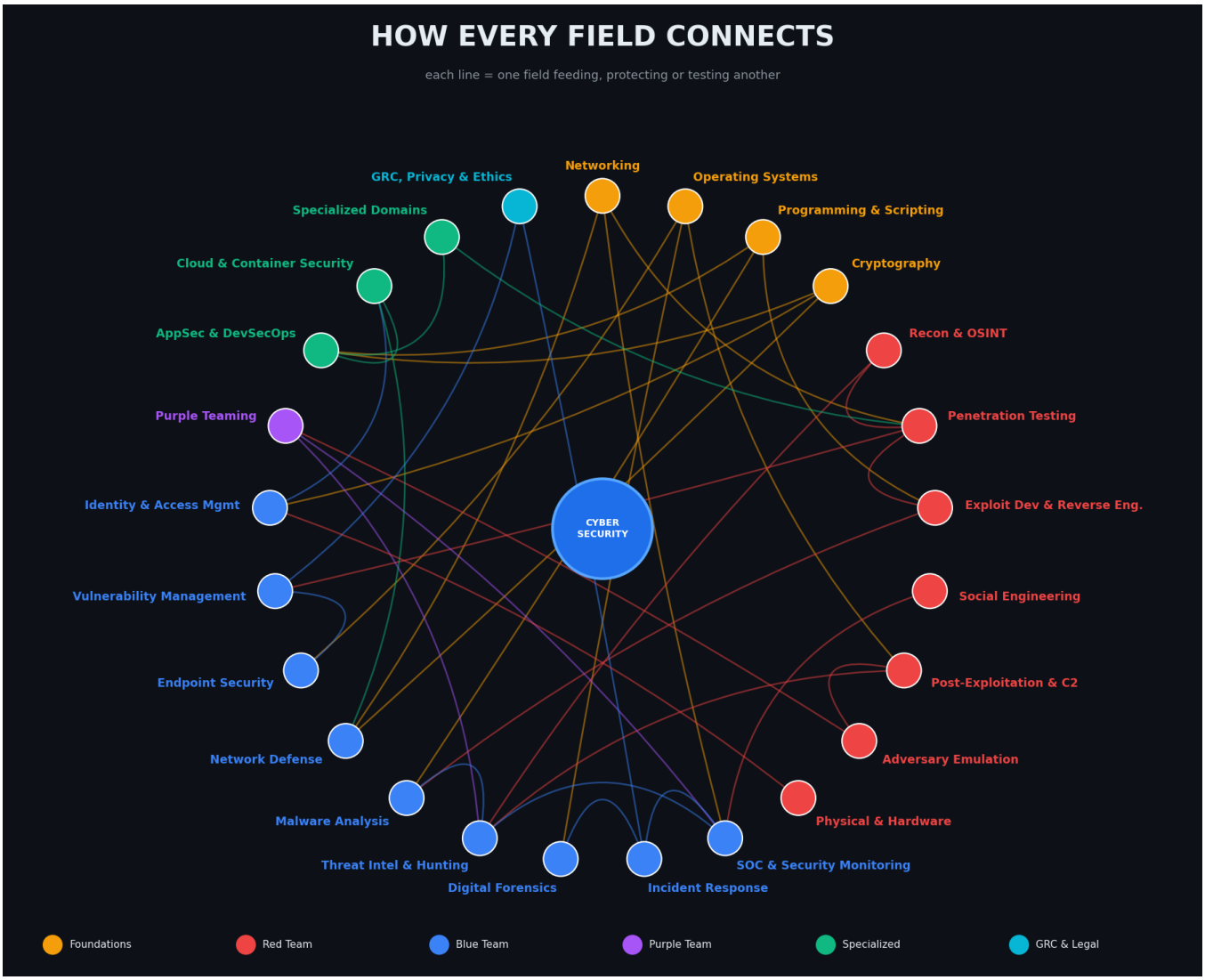
GRC, Privacy & Ethics `05_GRC_Legal/GRC_Privacy_Ethics/`

Policy, risk, audit, compliance and the law.

Sub-fields	Risk Management | Compliance (ISO 27001, NIST, PCI DSS, SOC 2) | Privacy (GDPR) | Audit & Cyber Law
Tools	Archer, OneTrust, Vanta
Certs	CISM, CISA, CISSP, ISO 27001 LA

8. How Everything Is Linked

Each line = one field feeding, protecting or testing another.



8.1 Link Table

From		To	Why they connect
Networking	->	Penetration Testing	protocol knowledge powers network attacks
Networking	->	Network Defense	firewalls & IDS need deep traffic knowledge
Networking	->	SOC & Security Monitoring	packet & log analysis
Operating Systems	->	Post-Exploitation & C2	privesc & persistence per OS
Operating Systems	->	Endpoint Security	hardening & EDR
Operating Systems	->	Digital Forensics	artifacts live inside the OS
Programming & Scripting	->	Exploit Dev & Reverse Eng.	write exploits & fuzzers
Programming & Scripting	->	AppSec & DevSecOps	code review & SAST
Programming & Scripting	->	Malware Analysis	tooling & reverse engineering
Cryptography	->	Identity & Access Mgmt	MFA, PKI, tokens
Cryptography	->	Network Defense	TLS, VPN, IPsec
Cryptography	->	AppSec & DevSecOps	secure storage & auth
Recon & OSINT	->	Penetration Testing	targets & attack surface
Recon & OSINT	->	Threat Intel & Hunting	OSINT overlaps with intel
Penetration Testing	->	Vulnerability Management	findings feed remediation
Penetration Testing	->	Exploit Dev & Reverse Eng.	deeper exploitation
Exploit Dev & Reverse Eng.	->	Malware Analysis	shared reverse engineering skills
Social Engineering	->	SOC & Security Monitoring	phishing detection
Post-Exploitation & C2	->	Threat Intel & Hunting	attacker TTPs become intel
Post-Exploitation & C2	->	Adversary Emulation	full attack chains
Adversary Emulation	->	Purple Teaming	attack simulation
Purple Teaming	->	SOC & Security Monitoring	new detections & tuning
Purple Teaming	->	Threat Intel & Hunting	ATT&CK coverage
SOC & Security Monitoring	->	Incident Response	alert becomes incident
Incident Response	->	Digital Forensics	evidence collection
Incident Response	->	GRC, Privacy & Ethics	lessons learned & reporting
Threat Intel & Hunting	->	SOC & Security Monitoring	IOCs & rules
Malware Analysis	->	Threat Intel & Hunting	YARA rules & IOCs
Vulnerability Management	->	GRC, Privacy & Ethics	risk register
Vulnerability Management	->	Endpoint Security	patching
Identity & Access Mgmt	->	Cloud & Container Security	cloud identity
AppSec & DevSecOps	->	Cloud & Container Security	DevSecOps pipelines
Cloud & Container Security	->	Network Defense	segmentation & zero trust
Physical & Hardware	->	Identity & Access Mgmt	badges & access control
Specialized Domains	->	Penetration Testing	IoT / mobile / OT testing
Specialized Domains	->	AppSec & DevSecOps	mobile & API security

9. Red vs Blue: The Kill Chain

Every attacker step has a defender answer. This is the heart of Purple Teaming.

Phase	Red Team does	Blue Team answers
1. Reconnaissance	OSINT, scanning, enumeration	Threat intel, honeypots, minimize exposure
2. Weaponization	Craft payloads & exploits	Malware analysis, YARA, sandboxing
3. Delivery	Phishing, drive-by, USB	Email gateway, awareness training, web filtering
4. Exploitation	Abuse a vulnerability	Patching, IPS, hardening, vuln management
5. Installation	Persistence, backdoors	EDR, app allow-listing, integrity monitoring
6. Command & Control	Beacon to C2 server	DNS sinkhole, egress filtering, NDR
7. Actions on Objectives	Exfiltration, ransomware	DLP, backups, incident response, forensics

Types of Security Work

Offensive (Red)	Break systems to find weaknesses before criminals do.
Defensive (Blue)	Monitor, detect, respond and recover.
Collaborative (Purple)	Turn red findings into blue detections.
Engineering (Green)	Design and build security into products and cloud.
Governance (Cyan)	Rules, risk, audit, law and compliance.
Foundations (Amber)	Networking, OS, code and crypto: the base of everything.

10. Folder Structure (your study repo)

Which field lives in which folder. Each leaf folder holds notes/, labs/, tools/ and cheatsheets/.

```
Cybersecurity/
├── 00_Foundations/   [Core Knowledge]
│   ├── Networking/
│   ├── Operating_Systems/
│   ├── Programming_Scripting/
│   └── Cryptography/
├── 01_Red_Team/   [Offensive]
│   ├── Recon_OSINT/
│   ├── Penetration_Testing/
│   ├── Exploit_Development/
│   ├── Social_Engineering/
│   ├── Post_Exploitation_C2/
│   ├── Adversary_Emulation/
│   └── Physical_Hardware/
├── 02_Blue_Team/   [Defensive]
│   ├── SOC_Monitoring/
│   ├── Incident_Response/
│   ├── Digital_Forensics/
│   ├── Threat_Intel_Hunting/
│   ├── Malware_Analysis/
│   ├── Network_Defense/
│   ├── Endpoint_Security/
│   ├── Vulnerability_Management/
│   └── Identity_Access/
├── 03_Purple_Team/   [Collaborative]
│   └── Purple_Teaming/
├── 04_Specialized_Engineering/   [Cross-cutting]
│   ├── AppSec_DevSecOps/
│   ├── Cloud_Security/
│   └── Specialized_Domains/
├── 05_GRC_Legal/   [Governance]
│   └── GRC_Privacy_Ethics/
├── 06_Certs_Roadmap/   # Study plans, exam notes, voucher tracking
├── 07_Labs_CTF/   # HTB, THM, PicoCTF write-ups, home-lab builds
├── 08_Tools/   # Cheat sheets, scripts, tool configs
└── 09_Resources/   # Books, courses, frameworks (ATT&CK, NIST, OWASP)
```

11. Learning & Certification Roadmap

Pick a track after the foundations. Most people start with Security+ or eJPT.

Track	Certifications
Beginner	Security+, CEH, eJPT, CC (ISC2), Google Cybersecurity
Red Path	PNPT, OSCP, CRTO, OSWE, OSED
Blue Path	BTL1, CySA+, GCIH, GCFA, SC-200
Management	CISSP, CISM, CISA, CCSP

Suggested order

1	**Foundations**	Networking, Linux, Python, Crypto basics
2	**Pick a side**	Red (pentest) or Blue (SOC) to start, then cross-train
3	**Practice**	TryHackMe, HackTheBox, CTFs, a home lab, write-ups on GitHub
4	**Specialize**	Cloud, AppSec, DFIR, malware, threat intel
5	**Credential**	Exam that matches your track